

Cloud Engineering Lab

Lab Completion Proof & Evidence Record

Lab: Linux VM Provisioning, Networking, Nginx & Python Validation

Assessment Area	Completion Evidence
Cloud VM Provisioning	Linux VM provisioned and running
Network Security	SSH (22) and HTTP (80) configured
Linux Administration	Package management and systemctl used
Web Server	Nginx installed, started and enabled
Internet Verification	Nginx welcome page verified using VM public IP
Python Scripting	Python script executed successfully against localhost:80
IaC Understanding	Reflection covering Terraform / CloudFormation automation

1. VM Provisioning Evidence

The lab requires a Linux virtual machine to be provisioned on a major cloud platform. The completed environment should show the VM in a Running state with a public IPv4 address.

Evidence Item	Expected Proof
Cloud platform	AWS EC2 / equivalent cloud VM
Operating system	Ubuntu Server / Linux
Instance state	Running
Public connectivity	Public IPv4 address assigned

Screenshot Evidence Placeholder: Insert the AWS console screenshot showing the VM as Running and the Public IPv4 address.

2. Network Security Evidence

The VM security group/firewall should permit SSH for administration and HTTP for the Nginx web service.

Rule	Protocol / Port	Recommended Source
SSH	TCP 22	My IP
HTTP	TCP 80	0.0.0.0/0

Screenshot Evidence Placeholder: Insert the security-group inbound-rule screenshot if required by the evaluator.

3. Nginx Deployment & Service Validation

Nginx was installed using the Ubuntu package manager and configured as a system service. The expected validation command is **sudo systemctl status nginx**, which should report **Active: active (running)**.

Validation Commands

```
sudo apt update
sudo apt install -y nginx
sudo systemctl start nginx
sudo systemctl enable nginx
sudo systemctl status nginx
```

Expected Result: Nginx service is active and running.

Screenshot Evidence Placeholder: Insert the SSH terminal screenshot showing **Active: active (running)**.

4. Internet Accessibility Validation

The Nginx default page should be reachable from a local browser through the VM's public IP address. This validates the path from the internet through the cloud firewall/security group to the VM and Nginx.

Expected Result: Browser displays the Nginx welcome page at `http://<VM-PUBLIC-IP>`.

Screenshot Evidence Placeholder: Insert the browser screenshot showing the Nginx welcome page and public IP in the address bar.

5. Python Automation Validation

A Python script named **check_nginx.py** uses Python's built-in HTTP client to send a GET request to localhost on port 80 and report the HTTP response.

Expected command:

```
python3 check_nginx.py
```

Expected output:

```
Nginx status: 200 OK
Nginx is running and responding.
```

Screenshot Evidence Placeholder: Insert the SSH terminal screenshot showing the Python command and successful 200 OK output.

6. IaC Reflection

Infrastructure as Code tools such as Terraform or AWS CloudFormation could automate the creation of the VM, networking configuration, security-group rules, and other infrastructure resources used in this lab. Nginx installation and configuration could also be automated using cloud-init, user data, or configuration-management tools. This would make deployments faster, repeatable, consistent, and less prone to manual errors. IaC configurations can also be stored in version control, reviewed, and reused across multiple environments.

7. Completion Checklist

✓	Linux VM provisioned and running
✓	Public IP assigned
✓	SSH connectivity verified
✓	HTTP port 80 configured
✓	Nginx installed and running
✓	Nginx accessible from the internet
✓	Python validation script executed
✓	IaC reflection completed

Important: This document is an evidence template/record for the completed lab. For submission as actual proof, attach or paste the real screenshots from your cloud console, browser, SSH terminal, and Python execution into the designated evidence sections.